

## **DRAWING DESCRIPTIONS — PATENT A**

### **Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems**

#### **BRIEF DESCRIPTION OF THE DRAWINGS**

**FIG. 1** is a system architecture block diagram showing the eight-layer self-sustaining neural-motor energy harvesting loop.

**FIG. 2** is a comparative energy balance chart showing control system energy depletion versus experimental system energy growth over 2000 operational steps.

**FIG. 3** is a schematic diagram of the spiking neural network architecture showing the leaky integrate-and-fire neuron model with synaptic connections.

**FIG. 4** is a circuit schematic of the energy harvesting subsystem comprising piezoelectric and thermoelectric transducers with rectification and storage.

**FIG. 5** is a graph showing the activity-dependent energy dynamics curve illustrating the quadratic cost model and optimal activity operating region.

**FIG. 6 is a hardware reference design layout diagram showing component placement and interconnections for the physical embodiment.**

**FIG. 7 is a validation results summary table confirming all falsifiable claims pass the specified thresholds.**

**FIG. 8 is a reference architecture diagram of the energy-bounded recursive control system showing the canonical signal flow from sensor interface through neuromorphic core, recursive reflection, energy governor, cognitive modulation, neural router, and actuation, with energy harvesting feedback.**

## **DETAILED DESCRIPTION OF THE DRAWINGS**

**FIG. 1 — System Architecture Block Diagram (8-Layer Loop)**

**This figure shows eight labeled blocks arranged in a vertical flow with arrows indicating signal direction, forming a closed loop.**

**Block layout (top to bottom, then looping back):**

**ENVIRONMENT (external, at top)**

|

**v**

[L1: Printed Membrane] — labeled "ThermochromicMixin"

|

**v**

[L2: Sensing Pads] — labeled "light\_intensity, membrane\_temp"

|

**v**

[L3: Optical Transmission] — labeled "signal\_voltage = light / 1000 x 5.0"

|

**v**

[L4: Neuromorphic CPU (SNN)] — labeled "BaseSNN (LIF neurons)"

| **Arrow entering from L8 labeled**

| **"reflection feedback"**

**v**

[L5: Servo Actuation] — labeled "target\_angle = clip(output x 180)"

|

+-----> [L6: Energy Harvesting] — labeled "piezo + thermal"

|

v

[L7: Ground Reference] — labeled "noise floor / baseline"

|

v

[L8: Recursive Reflection] — labeled "previous\_output"

|

+-----> (arrow loops back to L4)

**Three signal pathways should be distinguished by line style:**

**- Solid line: Main spine (L1 through L8 to L4)**

**- Dashed line: Thermal cross-link (L1 membrane\_temp directly to L6)**

- Dotted line: Reflection feedback (L8 to L4)

All blocks are rectangles with rounded corners. Environment is shown as a

cloud shape. The loop-back arrow from L8 to L4 should be prominent and labeled

"THE LOOP CLOSES HERE."

### **FIG. 2 — Energy Balance Comparison**

This figure shows a dual-panel line graph or single graph with two curves.

**X-axis: Operational Steps (0 to 2000)**

**Y-axis: Cumulative Energy (mWh)**

**Curve 1 (Control — solid line, labeled "EnergyConfig (Control)"):**

- Starts at 50.0 mWh

- Decreases linearly

- Ends at approximately -4,697 mWh at step 2000

- Label: "base\_consumption = 470 mW"

**Curve 2 (Experimental — dashed line, labeled "BalancedEnergyConfig (Experimental)"):**

- Starts at 50.0 mWh**
- Rises rapidly to 100.0 mWh (storage capacity) within approximately 30 steps**
- Plateaus at 100.0 mWh for remaining steps (homeostatic equilibrium)**
- Label: "base\_consumption = 45 mW, capacity = 100 mWh, overflow → heat"**

**A horizontal dashed line at  $y = 0$  labeled "Self-Sustaining Threshold."**

**A horizontal dashed line at  $y = 100$  labeled "Storage Capacity (100 mWh)."**

**Annotations:**

- Arrow pointing to control curve: "System fails — energy depleted"**
- Arrow pointing to experimental curve plateau: "System achieves homeostasis at capacity"**

**The area between the two curves should be lightly shaded to emphasize the divergence.**

### **FIG. 3 — SNN Architecture (LIF Neuron Model)**

This figure shows the internal structure of the spiking neural network.

#### **Top section — Single LIF Neuron Detail:**

- Circle labeled "Neuron i"
- Input arrows entering from left: " $I_{\text{external}}$ " and " $I_{\text{reflection}}$ "
- Inside the circle: equation written as " $V(t+1) = V(t) \times \text{leak} + I \times dt$ "
- Threshold line: horizontal dashed line labeled " $V_{\text{threshold}} = 0.5$ "
- Output arrow exiting right: "spike (0 or 1)"
- Below neuron: "If  $V \geq \text{threshold}$ : spike,  $V = V_{\text{reset}}$ "

#### **Bottom section — Network Topology:**

- N circles (labeled "1" through "N") arranged in a single layer
- All-to-all connections shown as lines between neurons
- Weight matrix notation: " $W[i,j]$ " on one representative connection

- Input arrow from left entering all neurons: "adjusted\_input"
- Output arrows from all neurons going right: "spike\_vector"
- Self-connections shown as small loops, crossed out (zeroed)
- Label: "N = 50 neurons (operational), N = 20 neurons (control)"

Inset box showing STDP rule:

- "If post fires:  $W += A+ \times \text{pre\_trace}$  (LTP)"
- "If pre fires:  $W -= A- \times \text{post\_trace}$  (LTD)"
- " $A- > A+$  (stability bias)"

#### FIG. 4 — Energy Harvesting Circuit

This figure shows an electrical circuit schematic.

Left side — Piezoelectric Harvester:

- Piezo disc symbol (capacitor with "PIEZO" label)
- Connected to servo shaft via mechanical coupling (zigzag line)



- Label: "27mm piezoelectric disc"
- Output leads to full-bridge rectifier (4 diodes in bridge configuration)
- Rectifier output to smoothing capacitor
- Label: "friction\_factor = 18.0"

**Center — LC Resonance Circuit:**

- Inductor symbol in series with piezo output
- Label: "Laird ferrite-backed inductor"
- Resonance annotation: " $f_{\text{res}} = 1 / (2 \times \pi \times \sqrt{L \times C})$ "
- Label: "3-10x voltage amplification at resonance"

**Right side — Thermoelectric Harvester:**

- Thermistor/thermocouple symbol
- Label: "NTC 10K 3950"
- Connected to separate rectifier path

- Label: "thermal\_factor = 8.0"

**Bottom — Energy Storage:**

- Both harvester outputs combine at a summing junction

- Arrow to storage element (battery/capacitor symbol)

- Label: "energy\_stored (mWh)"

- Output arrow labeled "Power to SNN (L4)"

**Power budget annotation box:**

- "Harvest = friction\_harvest + thermal\_harvest"

- "Cost = base\_consumption + activity\_cost x spike\_count^2"

- "Net = Harvest - Cost"

**FIG. 5 — Activity-Dependent Energy Dynamics**

This figure shows a graph derived from `BalancedEnergyConfig` parameters (core/energy.py) illustrating the energy harvesting and consumption curves as functions of neural spike count.

**X-axis: "Neural Activity (Spikes / Step)" ranging from 0 to 50**

**Y-axis: "Energy per Step (mWh)"**

**Two curves:**

**1. Harvest curve (ref 500, dashed line): Nearly linear, gently increasing**

**- Formula:  $\text{Harvest} = 0.20 + (0.09 \times \text{spikes}) \text{ mWh/step}$**

**- Composed of thermal baseline (0.20 mWh at  $\Delta T = 5$  degrees C) plus**

**piezoelectric harvest proportional to spike-driven motor activity**

**- Rises from 0.20 mWh at 0 spikes to approximately 4.7 mWh at 50 spikes**

**2. Cost curve (ref 502, solid line): Quadratic, starting near harvest level**

**- Formula:  $\text{Cost} = 0.225 + (0.006 \times s) + (0.0003 \times s^2) \text{ mWh/step}$**

**- Composed of base consumption ( $45 \text{ mW} \times 0.005 \text{ h} = 0.225 \text{ mWh}$ ) plus**

**linear activity cost ( $1.2 \text{ mW} \times 0.005 \text{ h} = 0.006 \text{ mWh/spike}$ ) plus**

**quadratic burst penalty ( $0.06 \times 0.005 = 0.0003 \text{ mWh/spike squared}$ )**

**- Rises from 0.225 mWh at 0 spikes to approximately 1.3 mWh at 50 spikes**

Crossover point (ref 506): Harvest equals Cost at approximately 0.3 spikes. Below this point (0 to 0.3 spikes), cost exceeds harvest (quiescent drain). Above this point (0.3 to 50+ spikes), harvest exceeds cost (net positive).

**Shaded region (ref 504, hatched): Area between curves where harvest exceeds**

**cost, labeled "NET POSITIVE REGION" with sublabel "Harvest >> Cost (Self-Sustaining)."**

Formula annotation box (ref 508): Contains exact formulas for both curves.

**Zone annotation at bottom: "Zone 1: Quiescent Drain — 0-1 Spikes: Cost > Harvest"**

Zero reference line: Horizontal dotted line at  $y = 0$ .

## **FIG. 6 — Hardware Reference Design**

This figure shows a top-down component layout diagram.

**Central component:**

**- Rectangle labeled "Raspberry Pi Pico (RP2040)" with pin labels on edges**

**Connected components (arranged around the Pico):**

**Top left:**

- Circle labeled "Piezo Disc 27mm"
- Connected to Pico via ADC pin (GP26/ADC0)
- Mechanical coupling line to Servo

**Top right:**

- Rectangle labeled "SG90 Micro Servo"
- Connected to Pico via PWM pin (GP0)
- Mechanical coupling to Piezo disc (zigzag line)

**Bottom left:**

- Small component labeled "NTC 10K Thermistor"
- Connected to Pico via ADC pin (GP27/ADC1)

**Bottom center:**

- Small component labeled "GL5528 LDR"

- Connected to Pico via ADC pin (GP28/ADC2)

**Bottom right:**

- Small square labeled "WS2812B RGB LED"
- Connected to Pico via digital pin (GP15)

**Left side:**

- Component labeled "Laird Ferrite Inductor"
- Connected in series between Piezo and rectifier
- Label: "LC resonance circuit"

**Power section:**

- Rectifier bridge symbol
- Storage capacitor
- Arrow labeled "Harvested energy feedback"

**BOM annotation box in corner:**

- "Raspberry Pi Pico (RP2040)"
- "SG90 Micro Servo"
- "27mm Piezoelectric Disc"
- "NTC 10K Thermistor"
- "Photoresistor (LDR)"
- "WS2812B RGB LED"
- "Misc (wires, board, inductor)"

#### FIG. 7 — Validation Results Summary

This figure shows a structured table summarizing all validated claims.

**Table with columns: Claim | Criterion | Measured Value | Threshold | Result**

**Section header: "Phase 7 Control Baseline (EnergyConfig)"**

**Row A: Closed-loop continuity | All channels 2000 points | 2000 | 2000 | PASS**

**Row B: Boundedness | All state variables bounded | bounded | bounded | PASS**

**Row C: Robustness under noise | Std(theta) <= 45 deg | [value] | 45 | PASS**

**Row D: Input-output gain | corr(L,theta) >= 0.2 | [value] | 0.2 | PASS**

**Row E: Saturation ratio | sat\_theta <= 0.20 | [value] | 0.20 | PASS**

**Section header: "Phase 7 Full Integration (Control confirms energy depletion)"**

**Row: Energy at 2000 steps | Energy depleted | -4,697 mWh | < 0 | CONFIRMED**

**Section header: "MCP Integration (BalancedEnergyConfig)"**

**Row: Energy at 2000 steps | Homeostasis at capacity | 100.0 mWh (capped) | > 0 |  
CONFIRMED**

**Section header: "Phase 8 STDP"**

**Row F8.1: Weight entropy decrease | final < initial | 2.95 to 2.02 bits | decrease | PASS**

**Row F8.2: STDP MI > 1.2x frozen | ratio | 6.52x | > 1.20 | PASS**

**Row F8.3: Late weight delta < 10% early | ratio | 0.00% | < 10% | PASS**

**Footer: "All claims validated. System is reproducible via git tags."**



**FIG. 8 — Energy-Bounded Recursive Control Architecture**

**This figure is the canonical embodiment and reference signal flow for the**

**entire system. It shows the complete architecture as a vertically stacked**

**block diagram with bidirectional control paths and energy feedback. This is**  
the primary reference figure for patent examination.

**System operation is constrained by real-time energy availability derived from**

**physical interaction with the environment, thereby ensuring bounded, non-abstract execution.**

**Layout — Vertically stacked functional blocks with lateral connections:**

## TOP — Physical Environment:



## ■ PHYSICAL ENVIRONMENT ■



**Two downward arrows from Environment:**

**Left arrow: "(Light / Temp / Force)" — to Sensor Interface**

**Right arrow: "(Mechanical Load)" — to Energy Harvesting**

**TIER 1 — Dual Input (left and right, same level):**

**Left block:**



**■ SENSOR INTERFACE ■**

**■ (BH1750, DS18B20, ■**

**■ Piezo Input) ■**



**Label: Specific sensor ICs for light (BH1750), temperature (DS18B20),**

**and force (piezo input)**

Arrow down to Neuromorphic Core

Right block:



■ ENERGY HARVESTING ■

■ (Piezo → Rectifier → ■

■ Supercap) ■



Label: Physical energy transduction chain: piezoelectric element through

full-bridge rectifier to supercapacitor storage

Arrow down to Neuromorphic Core

TIER 2 — Central Processing (full width):



■ NEUROMORPHIC CORE (SNN) ■



■ • Spiking Neurons ■

■ • Membrane Dynamics ■

■ • Optional STDP ■

■ • Activity-Dependent Energy Cost ■



Two downward arrows from Neuromorphic Core:

Left arrow: to Recursive Reflection

Right arrow: to Energy Governor

Internal labels:

- "Spiking Neurons" — leaky integrate-and-fire neural population

- "Membrane Dynamics" —  $V(t) = V(t-1) \times \text{leak} + I_{\text{adjusted}}$

- "Optional STDP" — spike-timing dependent plasticity (opt-in)

- "Activity-Dependent Energy Cost" — quadratic:  $\text{base} + k \times (\text{spikes}/N)^2$

**TIER 3 — Dual Processing (left and right, same level, BIDIRECTIONAL):**

**Left block:**

[illegible]

## ■ RECURSIVE REFLECTION ■



■ • Prior State Memory ■

## ■ • Output Feedback ■

## ■ • Temporal Context ■

**Label: Self-observation mechanism** — feeds aggregate output at timestep  $t$

**back as input at timestep t+1, scaled by reflection coefficient**

**Right block:**



Both blocks have arrows downward to Cognitive Modulation Layer

TIER 4 — Modulation (full width):



■ COGNITIVE MODULATION LAYER ■



■ • Attention Weighting ■

■ • Intention Bias ■

■ • Contextual Suppression / Amplification ■

■ • (Optional) External Advisory Input ■



Two downward arrows:

Left arrow: to Neural Router

## Right arrow: to Telemetry & Logging

**Internal labels:**

- **"Attention Weighting"** — prioritizes input channels
- **"Intention Bias"** — goal-directed modulation
- **"Contextual Suppression / Amplification"** — state-dependent gating
- **"(Optional) External Advisory Input"** — external control layer when connected

**TIER 5 — Output (left and right, same level):**

**Left block:**

[illegible]

## ■ NEURAL ROUTER ■



## ■ • Priority Arbitration■

## ■ • Energy Gating ■



\_\_\_\_\_

**priority, gates signals when energy is insufficient, prevents output**

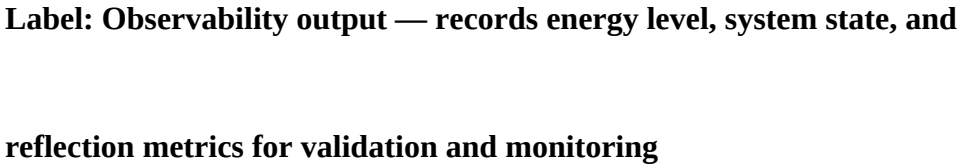
## Arrow down to Actuation & Response

[illegible]

■ ■

■ • State ■

## 25



### No downward arrow (terminal output)

### **TIER 6 — Actuation (full width):**



## ■ • Mechanical Interaction ■

## ■ • Environmental Perturbation ■



**Label: Physical output — servo control drives mechanical interaction,**

**which perturbs the environment (closing the loop back to Physical**

**Environment at top) and generates mechanical load (feeding Energy**

**Harvesting at Tier 1)**

**Two upward feedback arrows from Actuation:**

**1. Arrow back to PHYSICAL ENVIRONMENT (top): "Environmental Perturbation"**

**2. Arrow back to ENERGY HARVESTING (Tier 1 right): "Mechanical Load generates piezoelectric energy"**

**FEEDBACK LOOPS (mark each distinctly):**

**Loop A — Energy Harvesting Loop:**

**Actuation → Mechanical Load → Energy Harvesting → Neuromorphic Core**

**Style: Bold solid arrows**

**Label: "ENERGY LOOP — Motor activity generates power for neural computation"**

**Loop B — Recursive Reflection Loop:**

**Neuromorphic Core → Recursive Reflection → Neuromorphic Core**

**Style: Dotted arrows**

**Label: "REFLECTION LOOP — Prior state observation at configurable gain"**

**Loop C — Environmental Perturbation Loop:**

**Actuation → Environment → Sensor Interface → Neuromorphic Core**

**Style: Dashed arrows**

**Label: "SENSORY LOOP — Actions change environment, environment changes input"**

**Loop D — Energy Governor Feedback:**

**Energy Harvesting → Energy Governor ↔ Recursive Reflection → Cognitive Modulation**

**Style: Thin solid arrows**

**Label: "REGULATION LOOP — Energy availability constrains all processing"**

**Key annotations at bottom of figure:**

**"This architecture is physically grounded: all computation is constrained by real-time energy availability derived from the system's own mechanical interaction with its environment."**

**"No external power source is required for sustained operation under the balanced energy configuration."**

**FIG. 8 illustrates one exemplary embodiment; alternative embodiments may omit, combine, or substitute individual modules while preserving the energy-bounded recursive control loop.**

**Sheet numbering: "8/8" at top center**

**GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)**

**- Paper size: 8.5 x 11 inches (US Letter)**

- **Top margin: 1 inch**
- **Left margin: 1 inch**
- **Right margin: 5/8 inch**
- **Bottom margin: 3/8 inch**
- **Black lines on white background ONLY**
- **No color, no grayscale shading (use hatching patterns instead)**
- **Line weight: sufficiently heavy for adequate reproduction**
- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/8", "2/8", etc.
- **No frames or borders around drawing area**
- **All text in figures must be at least 1/8 inch (3.2 mm) high**
- **Reference numerals must be used consistently across all figures**